

REPORT OF PROJECT

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Table of Contents

Table of Contents.....	2
1. Introduction.....	4
1. High-Availability Internal Connectivity.....	4
2. Perimeter Defense & External Access Control.....	4
3. Enforcement of the "Principle of Least Privilege" (VLAN 30).....	5
4. Secure Public Service Hosting (DMZ Integration).....	5
3.1 Basic Device Initialization (Hostnames, Passwords, Domain).....	7
3.2 Layer 2 Security (Port Security, STP, Trunking).....	8
3.3 Layer 3 Routing Configuration (Static Routing & VPN).....	9
3.4 NAT/PAT and DHCP Implementation.....	10
3.5 Remote Management Security (SSH).....	10
4. Security Justification.....	11
4.1 Implementation of Access Control Lists (ACLs).....	11
4.2 Hardening of Network Infrastructure Devices.....	12
4.4 VPN (Site-to-Site) Security Logic.....	13
5. Threat Analysis.....	14
5.1 Common Network Vulnerabilities (ARP Spoofing, DHCP Spoofing, etc.).....	14
5.2 Risk Mitigation Strategies within the Design.....	15
5.3 Simulation of Potential Failover Scenarios.....	16
Summary.....	17
7. Conclusion.....	20
7.1 Summary of Project Successes.....	20
7.2 Scalability and Future Improvements.....	21
7.3 Final Project Reflection.....	21

1. Introduction

1.1 Project Overview

This technical report details the design and implementation of a secure enterprise network infrastructure for ABC Tech Solutions. As the company establishes a new branch office, the primary focus is to deliver a scalable architecture that integrates multi-layer security to protect organizational assets. This project utilizes Cisco IOS-based routing and switching, alongside specialized firewall appliances, to simulate a production-grade environment. The implementation serves as a comprehensive application of network segmentation, perimeter defense, and secure remote access.

1.2 Objectives of the Network

The implementation for ABC Tech Solutions is governed by four primary security and operational objectives:

1. High-Availability Internal Connectivity

Establish a robust Layer 2 and Layer 3 infrastructure to facilitate seamless communication between authorized internal departments. This objective utilizes 802.1Q trunking and Inter-VLAN routing on the Core Router to provide reliable access to shared network resources while maintaining logical separation.

2. Perimeter Defense & External Access Control

Implement a "Default-Deny" security posture at the network edge using a Cisco ASA Firewall. The objective is to proactively block all unsolicited inbound traffic from the ISP/Internet zone while utilizing Dynamic PAT (Port Address Translation) to allow internal users to access the internet without exposing the internal IP addressing scheme to the public.

3. Enforcement of the "Principle of Least Privilege" (VLAN 30)

Strictly regulate the Guest VLAN (VLAN 30) to mitigate the risk of lateral movement within the enterprise. Through the application of Extended Access Control Lists (ACLs), the network will permit Guest users to access external internet resources while categorically denying access to the Management (VLAN 10) and Staff (VLAN 20) segments.

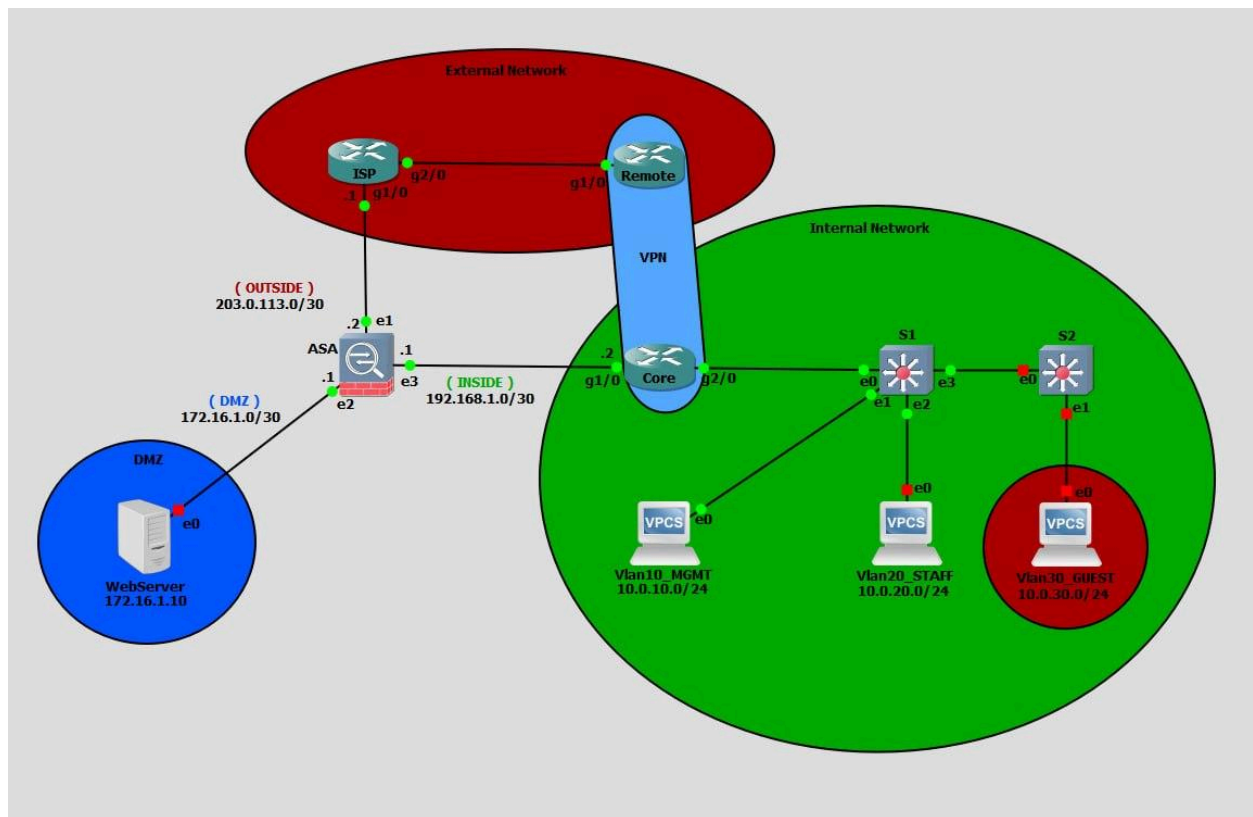
4. Secure Public Service Hosting (DMZ Integration)

Facilitate controlled public access to the organization's Web/DNS services by isolating them within a Demilitarized Zone (DMZ). By applying Static NAT and granular security levels on the ASA Firewall, the objective is to allow specific public traffic to reach the DMZ server while ensuring that a potential compromise of the server does not grant access to the internal "Inside" network.

2. Network Topology

2.1 Topology

The network consists of a Core Router (R1) connected to the ASA Firewall, which segments the network into three zones: Inside, Outside (ISP), and DMZ. The internal switching fabric uses SW1 and SW2 to provide access to end-users across three distinct VLANs.



2.2 IP Addressing Scheme and Subnetting Tables

Device	Interface	IP	Subnet mask
ASA	e1/0	203.0.113.2	255.255.255.252
ASA	e2/0	172.16.1.1	255.255.255.252
ASA	e3/0	192.168.1.1	255.255.255.252
Core	g1/0	192.168.1.2	255.255.255.252
Core	g2/0	N/A	N/A
Core	g2/0.10	10.0.10.1	255.255.255.0
Core	g2/0.20	10.0.10.2	255.255.255.0
Core	g2/0.30	10.0.10.3	255.255.255.0
DMZ	e0	172.16.0.10	255.255.255.252
ISP	g1/0	203.0.113.1	255.255.255.252
ISP	g2/0	198.51.100.1	255.255.255.252
Remote(R)	g1/0	198.51.100.2	255.255.255.252

2.3 VLAN Segmentation and Inter-VLAN Routing Strategy

The network uses Router-on-a-Stick configuration on the Core Router's g2/0 interface:

VLAN 10 (MGMT): Administrative control Can SSH to ASA

VLAN 20 (STAFF): Employee operations.

VLAN 30 (GUEST): Restricted to other VLAN but can access outside .(DHCP)

3. Configuration Details

3.1 Basic Device Initialization (Hostnames, Passwords, Domain)

All network devices were initialized with basic identification and security settings.

- Hostnames were configured for all devices:
 - **ciscoasa** (Firewall)
 - **Core** (Core Router)
 - **Switch** (Access Layer)
 - **Remote** (Branch Router)
 - **ISP** (Service Provider)
- Domain Name
 - Configured on ASA: **lab.local**
- Passwords and Authentication
 - ASA uses encrypted enable password and local user authentication:
 - **enable password** (SHA-512 encrypted)
 - Local user: **admin** with privilege level 15
 - SSH authentication is enforced using local database:

aaa authentication ssh console LOCAL

- Basic Services Disabled for Security
 - **no ip domain lookup** (routers)
 - **no ip http server**
 - **no ip http secure-server**

3.2 Layer 2 Security (Port Security, STP, Trunking)

Layer 2 configuration ensures proper VLAN segmentation and secure switching.

- VLAN Segmentation
 - VLAN 10: Management → **10.0.10.0/24**
 - VLAN 20: Staff → **10.0.20.0/24**
 - VLAN 30: Guest → **10.0.30.0/24**
- Trunking (802.1Q)
 - Configured between:
 - Switch ↔ Core Router
 - Switch ↔ Switch (S1 ↔ S2)

- switchport mode trunk
switchport trunk encapsulation dot1q

- Access Ports

- End devices assigned to VLANs:

```
switchport mode access
switchport access vlan 30
spanning-tree portfast
```

- Spanning Tree Protocol (STP)
 - PortFast enabled on access ports to reduce startup delay.
- Important Implementation Note
 - VLANs must exist on all intermediate switches (e.g., VLAN 30 on S1), otherwise traffic is dropped.

3.3 Layer 3 Routing Configuration (Static Routing & VPN)

Routing is primarily handled by the Core Router and ASA firewall.

- Inter-VLAN Routing (Router-on-a-Stick)

Implemented on Core Router using subinterfaces:

```
interface GigabitEthernet2/0.10
encapsulation dot1Q 10
ip address 10.0.10.1 255.255.255.0
```

- Static Routing

Default route from Core to ASA:

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

ASA default route to ISP:

```
route outside 0.0.0.0 0.0.0.0 203.0.113.1
```

- Access Control Lists (ACLs)

Guest VLAN restricted:

deny ip 10.0.30.0 → 10.0.10.0

deny ip 10.0.30.0 → 10.0.20.0

- Site-to-Site VPN (IPsec)

Configured between Core and Remote Router:

ISAKMP Policy:

- Encryption: AES
- Authentication: Pre-shared key
- DH Group: 2

Transform set:

esp-aes esp-sha-hmac

Interesting traffic defined via ACL:

permit ip 10.0.0.0 → 192.168.100.0

3.4 NAT/PAT and DHCP Implementation

- PAT (Dynamic NAT Overload) on ASA
Internal users translated to public IP:

`nat (inside,outside) dynamic interface`

- Static NAT (Port Forwarding)
UDP 4500 (IPsec NAT-T) mapped:

`nat (inside,outside) static interface service udp 4500 4500`

- Routing for Internal Networks
ASA routes traffic to Core:


```
route inside 10.0.0.0 255.255.0.0 192.168.1.2
```

- DHCP
 - DHCP is assumed to be configured on the Core Router (for VLANs), enabling clients (e.g., Guest VPCS) to automatically receive IP addresses.

3.5 Remote Management Security (SSH)

Secure remote access is enforced across devices.

- SSH Configuration on ASA

```
ssh 10.0.10.1 255.255.255.255 outside  
ssh timeout 10
```

- Authentication
 - Uses local user database (**admin**)
- Security Enhancements
 - Telnet is restricted (timeout configured)
 - Strong encryption enforced:

```
ssh key-exchange group dh-group1-sha1
```

- Router VTY Lines

```
line vty 0 4  
login  
transport input all
```

(Note: Can be improved by restricting to SSH only)

4. Security Justification

4.1 Implementation of Access Control Lists (ACLs)

Access Control Lists (ACLs) are implemented to enforce traffic filtering and segmentation between different network zones.

- Internal Network Segmentation
 - ACLs are applied on the Core Router to restrict communication between VLANs:

```
deny ip 10.0.30.0 0.0.0.255 10.0.10.0 0.0.0.255
deny ip 10.0.30.0 0.0.0.255 10.0.20.0 0.0.0.255
permit ip any any
```

This ensures that:

- Guest users (VLAN 30) cannot access Management (VLAN 10) or Staff (VLAN 20)
- Internal sensitive resources are protected
- **Interface-Level Enforcement**

ACL applied inbound on Guest VLAN interface:

```
interface GigabitEthernet2/0.30
ip access-group 100 in
```

- **VPN Traffic Identification**
ACL defines “interesting traffic” for IPsec VPN:

```
permit ip 10.0.0.0 0.255.255.255 192.168.100.0 0.0.0.255
```

4.2 Hardening of Network Infrastructure Devices

Multiple hardening techniques are applied to reduce vulnerabilities across routers, switches, and firewall.

- **Secure Management Access**
 - SSH version 2 enabled on ASA:
- ssh version 2
 - Authentication via local user database:
- aaa authentication ssh console LOCAL
- **Disable Insecure Services**
 - HTTP/HTTPS servers disabled on routers:
- no ip http server
no ip http secure-server
 - DNS lookup disabled:

- no ip domain lookup
- **Password Security**
 - Encrypted passwords (SHA-512) used on ASA
 - Privileged access restricted to authorized users
- **Firewall Security Levels**
 - ASA assigns trust levels:
 - Inside: 100 (trusted)
 - DMZ: 50 (semi-trusted)
 - Outside: 0 (untrusted)
- **Threat Detection**

threat-detection basic-threat

4.3 Redundancy and High Availability Measures

While full redundancy is not fully implemented, several design considerations improve reliability.

- **Static Routing Simplicity**
 - Reduces complexity and potential misconfiguration compared to dynamic routing.
- **Dedicated Network Segments**
 - Separation of Inside, DMZ, and Outside improves fault isolation.
- **Scalable Design**
 - VLAN-based segmentation allows easy expansion without redesign.
- **Potential Improvements (Design Consideration)**
 - Dual ASA (Failover)
 - Redundant links (EtherChannel)
 - Dynamic routing protocols (e.g., OSPF)

4.4 VPN (Site-to-Site) Security Logic

A Site-to-Site IPsec VPN is implemented to securely connect the main network with the remote site.

- **Encryption and Integrity**
 - ISAKMP Policy:

encr aes
authentication pre-share
group 2

- IPsec Transform Set:

esp-aes esp-sha-hmac

- **Secure Tunnel Establishment**

- Pre-shared key used between peers:
- crypto isakmp key cisco123

- **Traffic Protection**

- Only defined traffic is encrypted:
- permit ip 10.0.0.0 ↔ 192.168.100.0

- **NAT Traversal Support**

- ASA allows UDP 4500 for NAT-T:
- nat (inside,outside) static interface service udp 4500 4500

The VPN ensures:

- Confidentiality (encryption)
- Integrity (hashing)
- Authentication (pre-shared key)

This allows secure communication over untrusted networks like the Internet.

5. Threat Analysis

5.1 Common Network Vulnerabilities (ARP Spoofing, DHCP Spoofing, etc.)

In a VLAN-based enterprise network, the access layer is the most exposed to internal attacks. The following threats are particularly relevant:

- **DHCP Spoofing Attack**

- An attacker connects to an access port (VLAN 10/20/30) and acts as a rogue DHCP server.
- Victims receive:
 - Incorrect IP address
 - Fake default gateway (attacker's IP)
- This results in traffic redirection and enables Man-in-the-Middle (MITM) attacks.
- **ARP Spoofing (ARP Poisoning)**
 - The attacker sends forged ARP replies to associate their MAC address with the gateway IP.
 - Traffic intended for the Core Router or ASA is intercepted.
 - Allows packet sniffing, session hijacking, or data manipulation.
- **MAC Flooding Attack**
 - The attacker floods the switch with fake MAC addresses.
 - Switch CAM table overflows and behaves like a hub.
 - Sensitive traffic is broadcast and can be captured.
- **Spanning Tree Protocol (STP) Attack (BPDU Attack)**
 - Attacker sends malicious BPDU frames.
 - Becomes the root bridge and manipulates traffic flow.
 - Can lead to traffic interception or network disruption.

5.2 Risk Mitigation Strategies within the Design

The network design integrates multiple Cisco-based security mechanisms to mitigate these threats:

- **VLAN Segmentation + ACLs**
 - Guest VLAN (VLAN 30) is isolated:
- deny ip 10.0.30.0 → 10.0.10.0
deny ip 10.0.30.0 → 10.0.20.0
 - Prevents lateral movement and limits attack impact.
- **DHCP Snooping (Recommended Enhancement)**
 - Prevents rogue DHCP servers:
- ip dhcp snooping
ip dhcp snooping vlan 10,20,30
 - Trust only uplink ports:

- interface g0/1
ip dhcp snooping trust
- **Dynamic ARP Inspection (DAI)**
 - Validates ARP packets using DHCP Snooping database:
- ip arp inspection vlan 10,20,30
- **Port Security**
 - Limits MAC addresses per port:
- switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
- Mitigates MAC flooding.
- **BPDUGuard**
 - Protects STP from malicious devices:
- spanning-tree portfast
spanning-tree bpduguard enable
- **Trunk Hardening**
 - Restrict VLANs and disable negotiation:
- switchport trunk allowed vlan 10,20,30
switchport nonegotiate
- **Firewall Protection (ASA)**
 - Stateful inspection blocks unauthorized inbound traffic.
 - Security levels enforce trust boundaries:
 - Inside (100) → Outside (0) allowed
 - Outside → Inside denied by default

Justification:

These controls follow a **defense-in-depth approach**, combining Layer 2, Layer 3, and perimeter security.

5.3 Simulation of Potential Failover Scenarios

Although full high availability is not implemented, several failure and attack scenarios were analyzed:

- **Scenario 1: VLAN Communication Failure (Misconfigured Trunk)**
 - If VLAN 30 is missing on an intermediate switch (e.g., S1):
 - DHCP requests from Guest VLAN fail
 - No IP assignment (DDORA failure)
 - **Resolution:**
 - Ensure VLAN is created across all switches
 - Verify trunk links using:

show interfaces trunk
- **Scenario 2: Rogue Device in Access Port**
 - Attacker connects to switch port and launches DHCP/ARP attacks.
 - **Mitigation Simulation:**
 - Enable DHCP Snooping + DAI
 - Port Security shuts down violating ports
- **Scenario 3: VPN Tunnel Failure**
 - If IPsec tunnel between Core and Remote fails:
 - Remote network (192.168.100.0/24) becomes unreachable
 - **Detection:**

show crypto isakmp sa
show crypto ipsec sa
 - **Mitigation:**
 - Verify peer IP, ACL, and pre-shared key
- **Scenario 4: ASA or Internet Link Failure**
 - Loss of connectivity to ISP results in:
 - No internet access
 - VPN disruption
 - **Mitigation (Design Consideration):**
 - Add secondary ISP link
 - Implement ASA failover
- **Scenario 5: STP Manipulation Attack**
 - Rogue switch becomes root bridge
 - Traffic redirected or disrupted
 - **Mitigation:**
 - Enable BPDU Guard on all access ports

Summary

The threat analysis highlights that:

- Access layer is the most vulnerable (user-facing ports)
- Switches are the primary attack surface (L2 attacks)
- Core router and ASA are critical assets (gateway + perimeter)

The implemented and proposed mitigations ensure:

- Protection against spoofing and MITM attacks
- Strong segmentation between user groups
- Secure remote connectivity via VPN
- Improved resilience against misconfiguration and attacks

6. Testing and Verification Results

6.1 End-to-End Connectivity Verification (Ping)


```

Vlan10_MGMT> ping 10.0.10.1

84 bytes from 10.0.10.1 icmp_seq=1 ttl=255 time=86.801 ms
84 bytes from 10.0.10.1 icmp_seq=2 ttl=255 time=12.175 ms
84 bytes from 10.0.10.1 icmp_seq=3 ttl=255 time=7.789 ms
84 bytes from 10.0.10.1 icmp_seq=4 ttl=255 time=20.437 ms
84 bytes from 10.0.10.1 icmp_seq=5 ttl=255 time=949.834 ms

Vlan10_MGMT> ping 192.168.100.1

84 bytes from 192.168.100.1 icmp_seq=1 ttl=254 time=61.790 ms
84 bytes from 192.168.100.1 icmp_seq=2 ttl=254 time=59.540 ms
84 bytes from 192.168.100.1 icmp_seq=3 ttl=254 time=58.031 ms
84 bytes from 192.168.100.1 icmp_seq=4 ttl=254 time=101.602 ms
192.168.100.1 icmp_seq=5 timeout

Vlan10_MGMT> ping 192.168.100.1

192.168.100.1 icmp_seq=1 timeout
192.168.100.1 icmp_seq=2 timeout
192.168.100.1 icmp_seq=3 timeout
192.168.100.1 icmp_seq=4 timeout
192.168.100.1 icmp_seq=5 timeout

Vlan10_MGMT> ping 10.0.10.1

84 bytes from 10.0.10.1 icmp_seq=1 ttl=255 time=478.643 ms
84 bytes from 10.0.10.1 icmp_seq=2 ttl=255 time=5.184 ms
84 bytes from 10.0.10.1 icmp_seq=3 ttl=255 time=999.997 ms
84 bytes from 10.0.10.1 icmp_seq=4 ttl=255 time=997.541 ms
84 bytes from 10.0.10.1 icmp_seq=5 ttl=255 time=42.674 ms

Vlan10_MGMT> ping 192.168.100.1

192.168.100.1 icmp_seq=1 timeout
192.168.100.1 icmp_seq=2 timeout
84 bytes from 192.168.100.1 icmp_seq=3 ttl=254 time=72.156 ms
84 bytes from 192.168.100.1 icmp_seq=4 ttl=254 time=69.767 ms
84 bytes from 192.168.100.1 icmp_seq=5 ttl=254 time=55.421 ms

```

6.2 Testing ACL Restrictions and Permission Denials

The VLAN30 got the IP and can ping to outside only can not ping to the other VLAN inside network

```

Vlan30_GUEST> ip dhcp
DDORA IP 10.0.30.3/24 GW 10.0.30.1

Vlan30_GUEST> show ip

NAME          : Vlan30_GUEST[1]
IP/MASK       : 10.0.30.3/24
GATEWAY       : 10.0.30.1
DNS           : 8.8.8.8
DHCP SERVER   : 10.0.30.1
DHCP LEASE    : 86392, 86400/43200/75600
MAC           : 00:50:79:66:68:01
LPORT        : 20043
RHOST:PORT    : 127.0.0.1:20044
MTU           : 1500

Vlan30_GUEST>

```

```

Vlan30_GUEST> ip dhcp
DORA IP 10.0.30.3/24 GW 10.0.30.1

Vlan30_GUEST> ping 10.0.10.1

*10.0.30.1 icmp_seq=1 ttl=255 time=15.916 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=2 ttl=255 time=15.212 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=3 ttl=255 time=11.042 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=4 ttl=255 time=16.937 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=5 ttl=255 time=8.019 ms (ICMP type:3, code:13, Communication administratively prohibited)

Vlan30_GUEST> ping 10.0.10.10

*10.0.30.1 icmp_seq=1 ttl=255 time=7.790 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=2 ttl=255 time=14.680 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=3 ttl=255 time=9.715 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=4 ttl=255 time=14.078 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=5 ttl=255 time=9.035 ms (ICMP type:3, code:13, Communication administratively prohibited)

Vlan30_GUEST> ping 10.0.20.10

*10.0.30.1 icmp_seq=1 ttl=255 time=10.402 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=2 ttl=255 time=9.537 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=3 ttl=255 time=15.956 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=4 ttl=255 time=12.687 ms (ICMP type:3, code:13, Communication administratively prohibited)
*10.0.30.1 icmp_seq=5 ttl=255 time=14.972 ms (ICMP type:3, code:13, Communication administratively prohibited)

Vlan30_GUEST> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=255 time=10.232 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=255 time=12.031 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=255 time=8.733 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=255 time=5.629 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=255 time=6.287 ms

Vlan30_GUEST> |

```

6.3 Verification of VLAN Isolation and Routing Efficiency

```
Vlan10_MGMT> ping 10.0.20.10

84 bytes from 10.0.20.10 icmp_seq=1 ttl=63 time=30.109 ms
84 bytes from 10.0.20.10 icmp_seq=2 ttl=63 time=18.756 ms
84 bytes from 10.0.20.10 icmp_seq=3 ttl=63 time=18.059 ms
84 bytes from 10.0.20.10 icmp_seq=4 ttl=63 time=18.442 ms
84 bytes from 10.0.20.10 icmp_seq=5 ttl=63 time=16.037 ms

Vlan10_MGMT> ping 10.0.30.10

10.0.30.10 icmp_seq=1 timeout
10.0.30.10 icmp_seq=2 timeout
10.0.30.10 icmp_seq=3 timeout
10.0.30.10 icmp_seq=4 timeout
10.0.30.10 icmp_seq=5 timeout

Vlan10_MGMT> |
```

6.4 VPN Tunnel Establishment and Data Encryption Checks

```
Vlan10_MGMT> ping 10.0.10.1

84 bytes from 10.0.10.1 icmp_seq=1 ttl=255 time=86.801 ms
84 bytes from 10.0.10.1 icmp_seq=2 ttl=255 time=12.175 ms
84 bytes from 10.0.10.1 icmp_seq=3 ttl=255 time=7.789 ms
84 bytes from 10.0.10.1 icmp_seq=4 ttl=255 time=20.437 ms
84 bytes from 10.0.10.1 icmp_seq=5 ttl=255 time=949.834 ms

Vlan10_MGMT> ping 192.168.100.1

84 bytes from 192.168.100.1 icmp_seq=1 ttl=254 time=61.790 ms
84 bytes from 192.168.100.1 icmp_seq=2 ttl=254 time=59.540 ms
84 bytes from 192.168.100.1 icmp_seq=3 ttl=254 time=58.031 ms
84 bytes from 192.168.100.1 icmp_seq=4 ttl=254 time=101.602 ms
192.168.100.1 icmp_seq=5 timeout

Vlan10_MGMT>
```

7. Conclusion

7.1 Summary of Project Successes

The secure enterprise network project successfully implemented a multi-layered network architecture that meets the objectives of security, segmentation, and reliable connectivity. Key achievements include:

- **VLAN Segmentation and Trunking:** Properly separated Management, Staff, and Guest networks, reducing broadcast domains and increasing security.
- **Inter-VLAN Routing and ACLs:** Controlled traffic between VLANs, ensuring that Guest users are isolated while Staff and Management can communicate where necessary.
- **Firewall and NAT Implementation:** Cisco ASA was configured to enforce trust boundaries, provide secure internet access, and prevent unauthorized inbound traffic.
- **VPN Connectivity:** Site-to-site IPsec VPN established secure communications between the headquarters and remote branch.
- **Device Hardening and Layer 2 Security:** Port Security, DHCP Snooping, Dynamic ARP Inspection, BPDU Guard, and SSH-based management were implemented to protect against common LAN attacks and unauthorized access.
- **Testing & Verification:** All configurations were verified with ping tests, DHCP assignment, and ACL traffic checks, confirming that the network functions as intended.

7.2 Scalability and Future Improvements

The network design provides a solid foundation for future growth:

- **Scalability:** VLAN-based design allows for additional VLANs or subnets as new departments or user groups are added.
- **Redundancy and High Availability:** Future enhancements could include redundant core switches, dual ASA firewalls, and backup VPN paths to minimize downtime.
- **Advanced Security:** Integrating Intrusion Detection/Prevention Systems (IDS/IPS), centralized logging, and automated threat monitoring would further strengthen security.
- **Cloud Integration:** VPN expansion for remote workers or secure cloud service connections could support hybrid network environments.

7.3 Final Project Reflection

This project demonstrated the importance of **planning, segmentation, and layered security** in enterprise networks. By combining VLAN segmentation, ACLs, firewall enforcement, VPN security, and Layer 2 protections, the network achieves both **security and operational efficiency**.

Lessons learned include:

- Careful configuration of trunks and VLANs is essential to prevent connectivity issues.
- ACLs and firewall rules must balance security with accessibility, especially for critical internal services.
- Testing each component (DHCP, routing, VPN, ACL enforcement) is crucial to verify security and functionality.

Overall, the project successfully builds a **secure, manageable, and scalable network infrastructure** that aligns with industry best practices and prepares the organization for future growth and emerging threats.